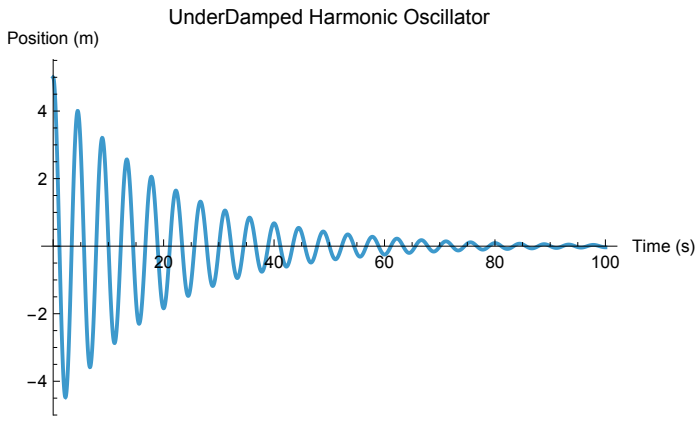
In[ ]:=
```
Clear[x];
m = 10;
k = 20;
b = 1;
x0 = 5;
v0 = 0;
sol =
 NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == 0, x[0] == x0, x'[0] == v0}, x, {t, 0, 100}]
Plot[Evaluate[x[t] /. sol], {t, 0, 100},
 PlotLabel → "UnderDamped Harmonic Oscillator",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All]
```

Out[ ]=

$$\{\{x \to \text{InterpolatingFunction}[\langle\text{Domain: } \{\{0., 100.\}\}\ \text{Output: scalar}\rangle]\}\}$$

Out[ ]=

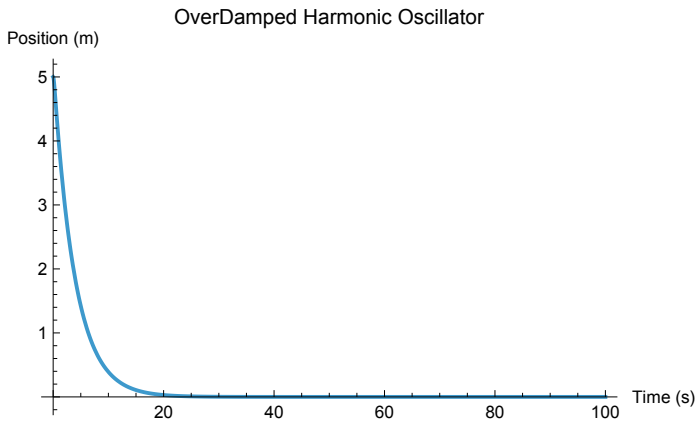


The damping parameters are below the critical threshold ($\gamma < 2\omega 0$). This causes the system to oscillate with an amplitude that decreases over time.

In[•]:=
```
Clear[x];
m = 10;
k = 20;
b = 80;
x0 = 5;
v0 = 0;
sol =
 NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == 0, x[0] == x0, x'[0] == v0}, x, {t, 0, 100}]
Plot[Evaluate[x[t] /. sol], {t, 0, 100},
 PlotLabel → "OverDamped Harmonic Oscillator",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All]
```

Out[•]=
```
{{x → InterpolatingFunction[ ⊞ Domain: {{0., 100.}}
                             Output: scalar ]}}
```

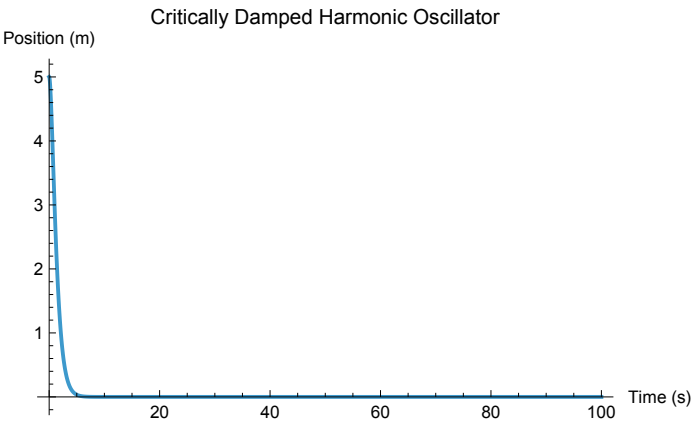
Out[•]=



In[•]:=
```
Clear[x];
m = 10;
k = 20;
b = 28.2842712474619;
x0 = 5;
v0 = 0;
sol =
 NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == 0, x[0] == x0, x'[0] == v0}, x, {t, 0, 100}]
Plot[Evaluate[x[t] /. sol], {t, 0, 100},
 PlotLabel → "Critically Damped Harmonic Oscillator",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All]
```

Out[•]=
```
{{x → InterpolatingFunction[ ⊞ Domain: {{0., 100. }}
                             Output: scalar ]}}
```

Out[•]=

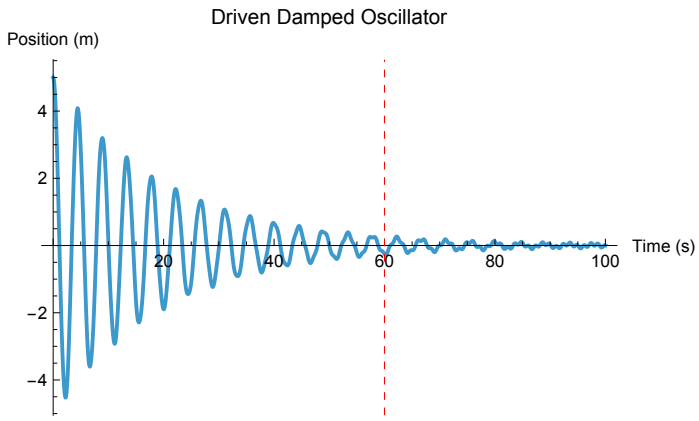


In[ ]:=
```
Clear[x, sol];
m = 10;
k = 20;
b = 1;
x0 = 5;
v0 = 0;
wd = 5;
F0 = 10;

sol = NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == (F0 / m) Cos[wd t],
     x[0] == x0, x'[0] == v0}, x, {t, 0, 100}];

Plot[Evaluate[x[t] /. sol], {t, 0, 100}, PlotLabel → "Driven Damped Oscillator",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All,
 Epilog → {Red, Dashed, InfiniteLine[{{60, 0}, {60, 1}}]}]
```

Out[ ]=



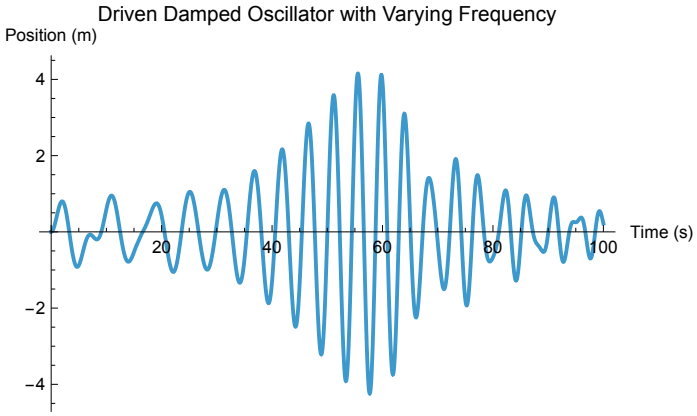
In[•]:=
```
Clear[x, sol];
m = 10;
k = 20;
b = 1;
F0 = 10;
x0 = 0;
v0 = 0;

sol = NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == (F0 / m) Cos[(0.5 + 0.01 * t) * t],
    x[0] == x0, x'[0] == v0}, x, {t, 0, 100}]

Plot[Evaluate[x[t] /. sol], {t, 0, 100},
 PlotLabel → "Driven Damped Oscillator with Varying Frequency",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All]
```

Out[•]=
```
{{x → InterpolatingFunction[ Domain : {{0., 100.}}
                             Output: scalar ]}}
```

Out[•]=



In[•]:=
```
Clear[x, sol];
m = 10;
k = 20;
b = 0;
F0 = 10;
wd = 1.3;
x0 = 0;
v0 = 0;

sol = NDSolve[{x''[t] + (b / m) x'[t] + (k / m) x[t] == (F0 / m) Cos[wd t],
    x[0] == x0, x'[0] == v0}, x, {t, 0, 100}]

Plot[Evaluate[x[t] /. sol], {t, 0, 100}, PlotLabel → "Driven Undamped Oscillator",
 AxesLabel → {"Time (s)", "Position (m)"}, PlotRange → All]
```

Out[•]=

$$\{\{x \to \text{InterpolatingFunction}[\langle\langle\text{Domain: } \{\{0.,\ 100.\}\}\ \text{Output: scalar}\rangle\rangle]\}\}$$

Out[•]=

